

19. Add slack variables to the following linear programming problem and write the initial simplex tableau:

$$\text{Maximize } Z = 3x_1 + 2x_2 + x_3$$

$$\text{subject to } 2x_1 + x_2 + x_3 \leq 150$$

$$2x_1 + 2x_2 + 8x_3 \leq 200$$

$$2x_1 + 3x_2 + x_3 \leq 320$$

$$\text{and } x_1, x_2, x_3 \geq 0.$$

(5 × 5 = 25 marks)

Section C

Answer any one question.

The question carries 11 marks.

20. Assume that the demand of an item A increases as the price decreases. If the weekly demand for A is q (in dollar \$) and the price is p then suppose that they are related by the linear relation $D(p) = 9 - 0.75q$.

- Find the quantity demanded at a price of \$ 5.25 per item and at a price of \$ 3.75 per item.
- It is also noticed that the quantity of item A supplied decreased as the price decreased. If price p and supply q are related by the linear function $S(p) = 0.75q$, find the quantity supplied at a price of \$ 5.25 per item and at a price of \$ 3.00 per item.
- Graph both functions $D(p)$ and $S(p)$ on the same axes.

21. Solve the problem using simplex method :

$$\text{Maximize } Z = 12x_1 + 15x_2 + 5x_3$$

$$\text{subject to } 2x_1 + 2x_2 + x_3 \leq 8$$

$$x_1 + 4x_2 + 3x_3 \leq 12$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0.$$

(1 × 11 = 11 marks)

FIFTH SEMESTER U.G. DEGREE EXAMINATION, NOVEMBER 2021

(CBCSS—UG)

Mathematics

MTS 5D 02—DISCRETE MATHEMATICS FOR BASIC AND APPLIED SCIENCES

(2019 Admissions)

Time : Two Hours

Maximum : 60 Marks

Section A

*Answer at least eight questions.
Each question carries 3 marks.
All questions can be attended.
Overall Ceiling 24.*

1. Define a proposition.
2. What is disjunction of two propositions ? Explain using its truth table.
3. Show that $p \wedge \bar{p}$ is a contradiction.
4. When do we say that two propositions are logically equivalent ?
5. Define binary operation.
6. Give example of an abelian monoid.
7. Give example of a cyclic group.
8. Give example of a semi group.
9. Define regular graph.
10. Define a complete bipartite graph.
11. When do we say that a graph is connected ?
12. Define a Hamiltonian graph.

(8 × 3 = 24 marks)

Section B

*Answer at least five questions.
Each question carries 5 marks.
All questions can be attended.
Overall Ceiling 25.*

13. Construct truth table for the compound proposition $(p \rightarrow q) \wedge (q \rightarrow p)$.
14. Show that $\overline{(p \wedge q)}$ and $\bar{p} \vee \bar{q}$ are logically equivalent.

Turn over

15. For an associative binary operation on a set S which has identity element, show that inverse if it exists is unique.
16. Show that the group $(\mathbb{Z}, +)$ is cyclic with generator 1 ?
17. For a Boolean algebra state and prove De Morgan's laws ?
18. Draw diagrams to represent the complete graph K_2 and the complete bipartite graph $K_{2,5}$.
19. Draw the diagram of sequence (2, 2, 2, 4, 4, 5, 5).

(5 × 5 = 25 marks)

Section C

*Answer any one question.
The question carries 11 marks.*

20. Describe the degree sequence of :
 - (a) The null graph with n vertices.
 - (b) The complete graph K_n .
 - (c) An r -regular graph with n vertices.
 - (d) The complete bipartite graph $K_{n,m}$ where $n \leq m$.
21. Construct truth tables for :
 - (a) $\neg(p \rightarrow q) \rightarrow \neg p$.
 - (b) $q \leftrightarrow (\neg p \vee \neg q)$.

(1 × 11 = 11 marks)